

Electricity Consumption Forecast with Artificial Neural Networks

Artificial Neural Networks 2025/2026

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Problem Statement

Electricity demand forecasting is essential for energy planning, grid stability, and cost optimization. In this exercise, you will design, train, and evaluate a neural network capable of predicting 12 months of future electricity consumption using historical data sampled every 15 minutes from 2011 to 2014.

You are provided with two datasets:

- **electricity_train.csv**: historical consumption for training (2011–2013).
- **electricity_test.csv**: consumption data for the evaluation period (2014).

Each file contains:

- **timestamp**: Date and time (15-min intervals)
- **consumption**: Normalized electricity consumption

The goal is to build a sequence-based forecasting model using artificial neural networks.

Operational Evaluation

Q1. Create a Neural Network with 1 Linear Layer - We begin with a minimal regression model that uses the last W steps to predict the next step. Note, Build an MLP with a single Linear layer that takes a flattened window and predicts the next 15-minute value.

Q2. Add Two Hidden Layers - Increase capacity with hidden layers to capture non-linear relationships. Note, extend the model to two hidden layers. No activations yet (that's next).

Q3. Add Activation Function - Non-linear activations increase expressive power. Note, Insert ReLU after each hidden layer.

Q4. Forward Pass Regression (Loss & One Step) - We want a scalar training objective to minimize prediction error. Note, Use MSELoss and verify the forward/backward path on a single batch.

Q5. Train the Neural Network - Fit the model on training windows for several epochs. Note, Train for 5 epochs and record average train loss per epoch.

Q6. Evaluate the Neural Network - We need simple, interpretable metrics. Note, Compute MAE, RMSE, MAPE on the test windows (horizon=1).

Q7. Hyperparameter Tuning (Tiny Grid) - Even small grids find better trade-offs of width/learning rate/batch size. Run a tiny grid over $\text{hidden1} \in \{64, 128\}$, $\text{hidden2} \in \{32, 64\}$, $\text{lr} \in \{1\text{e-}3, 3\text{e-}4\}$, $\text{batch} \in \{256, 512\}$; train 3 epochs each; keep the best MAE on test.

Q8. Plot Actual vs. Predicted on the Test Set (2014) - Before deploying a forecaster, teams always generate a visual sanity check of predictions against the held-out test year. Note, Plot Actual vs. Predicted for the 2014 test period at (a) 15-minute resolution for a single week and (b) hourly resolution for the full year.

Q9. Provide Forecast for the Next 12 Months - Deliver a full-year forecast (the entire 2015) at 15-minute granularity, using recursive one-step predictions. Note, Use your best tuned model to roll forward across the whole test horizon and produce a 12-month forecast series.

Q10. Plot the Forecast of 2015 (Out-of-Sample Roll-Forward) - Planners often need a next-year forecast even when no ground truth is available yet. Note, Using your trained model, generate a synthetic forecast for 2015 by recursive one-step predictions.